

Cyclistic Bike-Share Case Study

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Role: Data Analyst

Date: May 2025

**Tools Used: File Explorer (Windows), BigQuery,
Google Sheets**

1. Business Task

Cyclistic's marketing team aims to increase annual memberships by understanding how casual riders and annual members use the service differently. As a junior data analyst, I analyzed six months of historical ride data to identify usage patterns and behavioral trends that can inform strategies for converting casual riders into annual members.

2. Data Sources

- **Data Source:** Divvy trip data (used for Cyclistic case study)
 - **Months Analyzed:** November 2024 to April 2025
 - **Tools Used:**
 - Local File Explorer (to manage and unzip .csv files)
 - BigQuery (for cleaning and analyzing large datasets)
 - Google Sheets (for visualizing exported query results)
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3. Data Cleaning and Preparation

- Loaded six monthly `.csv` files into File Explorer and imported them into BigQuery.
 - Combined all data using `UNION ALL` in SQL.
 - Removed any rows with null or invalid timestamps (e.g., where `ended_at < started_at`).
 - Created a `ride_length_min` column using `TIMESTAMP_DIFF(ended_at, started_at, MINUTE)`.
 - Extracted `day_of_week` using `FORMAT_TIMESTAMP('%A', started_at)`.
 - Saved a cleaned and combined view in BigQuery for all further analysis.
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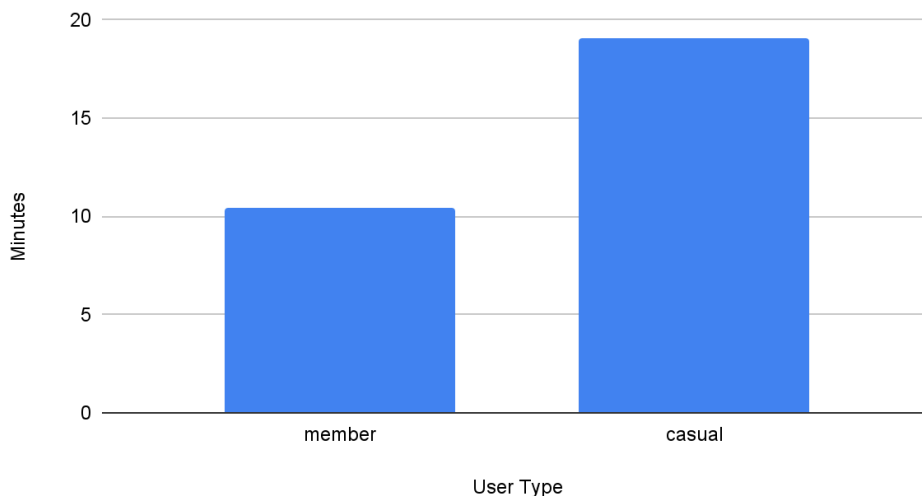
4. Analysis Summary

Using SQL, I discovered several key behavioral differences between casual riders and annual members:

- **Ride Duration:** Casual riders had a significantly higher average ride duration (~19 minutes) than members (~10 minutes).
 - **Weekly Usage Patterns:** Casual riders used the service more heavily on weekends, particularly on Saturdays and Sundays. In contrast, annual members showed more consistent weekday usage, likely for commuting.
 - **Daily Ride Trends:** Casual riders' ride lengths spiked on weekends, while members maintained steady ride times throughout the week.
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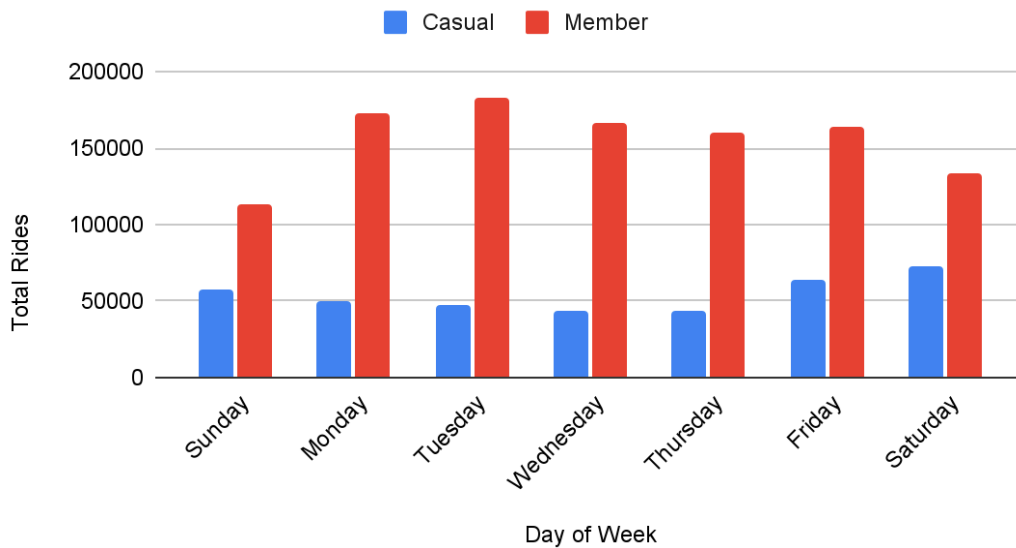
5. Supporting Visualizations

Average Ride Length by User Type (Minutes)



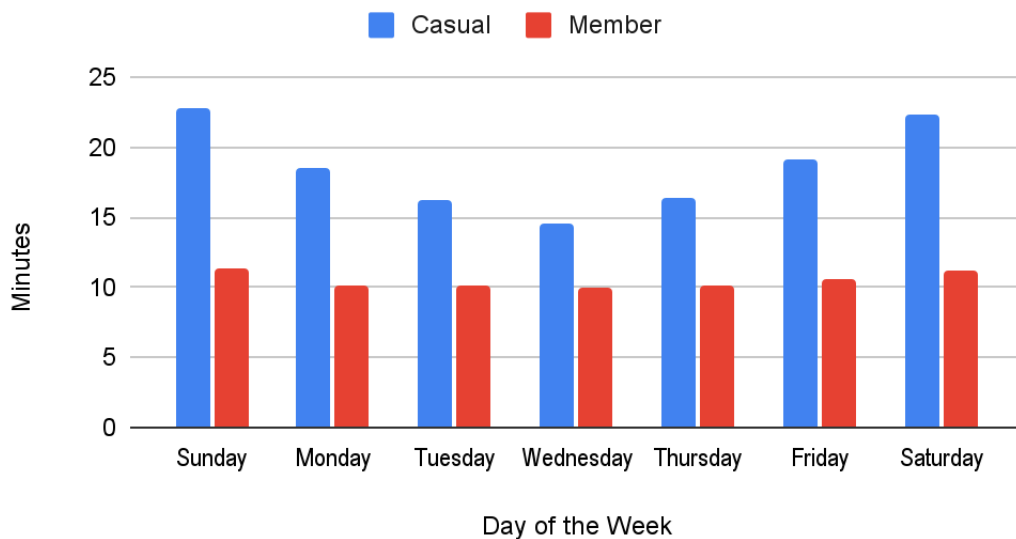
This chart shows that casual riders tend to ride longer than annual members.

Total Rides by Day of Week and User Type



Casual riders are most active on weekends, whereas annual members ride more during weekdays — consistent with commuting patterns.

Average Ride Length by Day of Week and User Type



Casual riders ride longer on weekends, while members have more balanced weekly ride lengths.

6. Top 3 Recommendations

1. **Weekend Membership Campaign:** Offer targeted membership discounts to casual riders on weekends, when their usage is highest.
2. **Commuter Incentives:** Promote benefits of annual membership (cost savings, flexibility) to frequent weekday casual riders who may be commuting.
3. **Digital Retargeting Campaigns:** Use email, app notifications, or ads to reach casual riders with messaging that aligns with their specific usage patterns — leisure vs. commuting.